

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.05.09.

1 Find the Reciprocal

Write the reciprocal of each number.

$$\frac{2}{3} \rightarrow \underline{\hspace{2cm}} \quad | \quad \frac{5}{8} \rightarrow \underline{\hspace{2cm}} \quad | \quad 4 \rightarrow \underline{\hspace{2cm}} \quad | \quad 1\frac{1}{2} \rightarrow \underline{\hspace{2cm}}$$

Answers _____

2 Reciprocal $\times$ Original = 1

Verify that each fraction times its reciprocal equals 1.

$$\frac{3}{7} \times \frac{7}{3} = \underline{\hspace{2cm}} \quad | \quad \frac{5}{9} \times \frac{9}{5} = \underline{\hspace{2cm}} \quad | \quad \frac{1}{4} \times \frac{4}{1} = \underline{\hspace{2cm}}$$

Answers _____

3 Find the Missing Factor

Each equation equals 1. Find the missing fraction.

$$\frac{4}{5} \times \underline{\hspace{2cm}} = 1 \quad | \quad \underline{\hspace{2cm}} \times \frac{7}{3} = 1 \quad | \quad 9 \times \underline{\hspace{2cm}} = 1$$

Answers _____

4 Reciprocal Patterns

Fill in the table.

What is the reciprocal of 0? Explain why it is undefined.

Table and Explanation _____

Fraction: $\frac{1}{3}$ | Reciprocal: _____ | Product: _____

Fraction: $\frac{2}{5}$ | Reciprocal: _____ | Product: _____

Fraction: $\frac{7}{4}$ | Reciprocal: _____ | Product: _____

Fraction: 1 | Reciprocal: _____ | Product: _____

5 Predict the Reciprocal

Kai says: "The reciprocal of a fraction greater than 1 is always less than 1, and the reciprocal of a fraction less than 1 is always greater than 1."

Test Kai's claim using $\frac{5}{3}$ and $\frac{3}{8}$. Is Kai correct? Explain using the definition of a reciprocal.

Explanation _____